

#Radio, Communications, Cellular Data, SIM Cards, Hardware

▣ **The ARRL Handbook for Radio Communications**

▣ **Digital Communications – John G. Proakis**

▣ **Understanding Digital Signal Processing – Richard Lyons**

▣ **Mobile Communications – Jochen Schiller**

▣ **LTE – The UMTS Long Term Evolution – Dahlman et al.**

▣ **Smart Card Handbook – Wolfgang Rankl & Wolfgang Effing**

 **Security of Mobile Communications – Jakobsson & Wetzel**

 **Making Embedded Systems – Elecia White**

 **Designing Embedded Hardware – John Catsoulis**

#Dev Ops

How Linux Works

UNIX and Linux System Administration Handbook

Computer Networking a Top Down Approach

TCP/IP Illustrated

The Linux Commandline

Classic Shell Scripting

ProGit

Contagious Delivery

Docker Deep Dive

Kubernetes Up and Running

Terraform: Up and Running

Designing Data Intensive Applications

Site Reliability Engineering

#OS and Interpreter and compiler and electronics and microprocessors and Storage devices

- **Operating Systems: Three Easy Pieces — Remzi & Andrea Arpaci-Dusseau**
- **Operating Systems: From 0 to 1 — Tu Do Hoang**
- **Linux Kernel Development — Robert Love**
- **Crafting Interpreters — Robert Nystrom**

- **Virtual Machines — Iain D. Craig**
- **Engineering a Compiler — Keith Cooper & Linda Torczon**
- **Modern Compiler Implementation in C — Andrew Appel**
- **Practical Electronics for Inventors — Paul Scherz & Simon Monk**
- **The Art of Electronics — Paul Horowitz & Winfield Hill**
- **(Optional) Learning the Art of Electronics — Hayes & Horowitz**
- **Computer Organization and Design (RISC-V Edition) — Patterson & Hennessy**
- **Structured Computer Organization — Andrew S. Tanenbaum**
- **Hard Disk Drive: Mechatronics and Control — W. Y. Lee**
- **Inside NAND Flash Memory — Kiyoun Kang**

#IOT

- **“Computer Organization and Design” — David A. Patterson & John L. Hennessy**

- **“Computer Networking: A Top-Down Approach” — Kurose & Ross**
- **“Programming Embedded Systems in C and C++” — Michael Barr**
- **“Embedded C Programming and the Atmel AVR” — Richard H. Barnett**
- **“Real-Time Concepts for Embedded Systems” — Qing Li**
- **“C++ Concurrency in Action” — Anthony Williams**
- **“Designing Data-Intensive Applications” — Martin Kleppmann (for general backend thinking)**
- **“IoT Security: Advances in Authentication” — Online resources / papers**
- **“Embedded Systems Firmware Demystified” — Ed Sutter**
- **“The C Programming Language” — Kernighan & Ritchie**
- **“C in Depth” / “Expert C Programming” — Peter van der Linden**
- **“C++ Primer” — Lippman, Lajoie, Moo**
- **“Effective Modern C++” — Scott Meyers**

- **“Making Embedded Systems” — Elecia White**
- **“Exploring Arduino” — Jeremy Blum**
- **“Programming the ARM Cortex-M4 in C” — Jonathan Valvano**
- **Optional: “Programming Embedded Systems in Assembly” (targeting your MCU architecture)**
- **“TCP/IP Illustrated” — W. Richard Stevens (read the networking fundamentals chapters)**
- **“Embedded Networking with TCP/IP” — Jan Axelson**
- **“MQTT Essentials” — Online book/resource**
- **“Practical IoT Security” — Brian Russell & Drew Van Duren**
- **“Real-Time Operating Systems Book 1: The Theory” — Jim Cooling**
- **“IoT Fundamentals: Networking Technologies, Protocols, and Use Cases” — David Hanes**
- **“C++ Templates: The Complete Guide” — Vandevoorde, Josuttis, Gregor**
- **“Effective Embedded C++” — Michael Barr**

- **TinyML: “TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers”**
- **“IoT Security Issues” — Academic papers + practical guides**

#Networks Internet Shutdown

- 1) Computer Networking: A Top-Down Approach — Kurose/Ross**
- 2) TCP/IP Illustrated, Volume 1 — The Protocols — W. Richard Stevens**
- 3) Routing TCP/IP (Vol 1 & 2) — Jeff Doyle & Jennifer Carroll**
- 4) Network Warrior — Gary A. Donahue**
- 5) Deep Packet Inspection: Technology and Practice — by Robert Birke**

6) Traffic Engineering with MPLS — Eric D. Osborne & Ajay Simha

****7) Practical Packet Analysis — Chris Sanders**

****8) Network Programmability and Automation — Jason Edelman, et al.**

****9) UNIX Network Programming — W. Richard Stevens**

#Web (Full Stack) - JavaScript

Book: “JavaScript: The Definitive Guide” — David Flanagan

Book: “You Don’t Know JS” series — Kyle Simpson

Book: “JavaScript and JQuery: Interactive Front-End Web Development” — Jon Duckett

Book: “Node.js Design Patterns” — Mario Casciaro & Luciano Mammino

• “Effective JavaScript” — David Herman

- **“Designing Data-Intensive Applications” — Kleppmann**

#Backend knowledge based on personal understanding (language agnostic)

- 1. “HTTP: The Definitive Guide” — David Gourley**
- 2. “Designing Web APIs” — Brenda Jin et al.**
- 3. “Flask Web Development” — Miguel Grinberg**
- 4. “Seven Databases in Seven Weeks” — Eric Redmond**
- 5. “Full Stack Development with React & Node” (book, not course)**
- 6. “Network Programming with POSIX Sockets”
“UNIX Network Programming” — Stevens**

- **MySQL Cookbook**
- **SQL: The Complete Reference**
- **High Performance MySQL**

#Backend (PHP)

📖 **“PHP & MySQL: Server-Side Web Development” — Jon Duckett**

📖 **“Modern PHP” — Josh Lockhart**

📖 **“PHP & MySQL Web Development” — Welling & Thomson**

📖 **“Laravel: Up & Running” — Matt Stauffer**

📖 **“Designing Web APIs” — Brenda Jin et al**

📖 **“HTTP: The Definitive Guide” — Gourley**

#Backend CSharp

Book: “C# 11 in a Nutshell” — Joseph Albahari & Ben Albahari

Book: “Programming C# 11” — Ian Griffiths

Book: “Pro ASP.NET Core 8” — Adam Freeman

Book: “Entity Framework Core in Action” — Jon P Smith

- **“Clean Architecture” — Robert C. Martin**

- **“Designing Data-Intensive Applications” — Martin**

Kleppmann

#Backend Java

Book: “Head First Java” — Kathy Sierra & Bert Bates

Book: “Effective Java” — Joshua Bloch

Book: “Spring in Action” — Craig Walls

**Book: “Java Persistence with Hibernate” — Christian
Bauer & Gavin King**

- **“Clean Code” — Robert C. Martin**

- **“Designing Data-Intensive Applications” — Martin**

Kleppmann

#Backend Python

Book: “Python Crash Course” — Eric Matthes

Book: “Effective Python” — Brett Slatkin

Book: “Flask Web Development” — Miguel Grinberg

Book: “Python and SQLAlchemy” (or equivalent ORM-focused book)

- **Alternative: Use Duckett-style exercises in SQLite / PostgreSQL with Python**
- **“Designing Data-Intensive Applications” — Martin Kleppmann (applies to Python systems too)**
- **“Fluent Python” — Luciano Ramalho**

#Client App Programming Windows C/C++ (SAMP level)

**** “C Programming: A Modern Approach” — K. N. King**

**** “C++ Primer” — Stanley B. Lippman et al.**

**** “Advanced Programming in the UNIX Environment” —
W. Richard Stevens**

**** “Windows System Programming” — Johnson M. Hart**

**** “Unix Network Programming, Volume 1: The Sockets
Networking API” — W. Richard Stevens**

**** “Network Programming for Microsoft Windows” —
Anthony Jones & Jim Ohlund**

**** “Programming with POSIX Threads” — David R. Butenhof**

**** “Windows via C/C++” — Jeffrey Richter**

**** “TCP/IP Sockets in C: Practical Guide for Programmers” — Michael J. Donahoo & Kenneth L. Calvert**

**** “Effective C++” + “More Effective C++” — Scott Meyers***

**** “Computer Networking: A Top-Down Approach” — Kurose & Ross**

#Cloud

Cloud foundations Pre programming

Book: “Cloud Computing: Concepts, Technology & Architecture” — Thomas Erl

**Book: “Cloud Computing: Principles and Paradigms” —
Rajkumar Buyya et al.**

**Book: “Microsoft Azure Essentials: Fundamentals of
Azure” — Michael S. Collier & Robin E. Shahan**

Book: “Cloud Architecture Patterns” — Bill Wilder

Cloud Programming (Without prior cloud knowledge)(CSharp)

**Book: “Cloud Computing: Concepts, Technology &
Architecture” — Thomas Erl**

**Book: “Cloud Computing: Principles and Paradigms” —
Rajkumar Buyya et al.**

**Book: “Microsoft Azure Essentials: Fundamentals of
Azure” — Michael S. Collier & Robin E. Shahan**

Book: “Cloud Architecture Patterns” — Bill Wilder

Book: “C# 11 in a Nutshell” — Joseph Albahari & Ben Albahari

Book: “Pro ASP.NET Core 8” — Adam Freeman

Book: “Microsoft Azure for Developers: Using C# and .NET” — John Paul Mueller

Book: “Hands-On Microservices with .NET 6” — Gaurav Arora & Ed Price

• **“Cloud Design Patterns: Prescriptive Architecture Guidance for Cloud Applications” — Microsoft Patterns & Practices**

Cloud Programming

Java

1. Effective Java – Joshua Bloch

2. Java Concurrency in Practice – Brian Goetz

1. Spring in Action – Craig Walls

2. Designing Data-Intensive Applications – Martin Kleppmann

1. Docker Deep Dive – Nigel Poulton

2. How Linux Works – Brian Ward

1. Kubernetes Up & Running – Kelsey Hightower

2. Cloud Native Java – Josh Long & Kenny Bastani

• AWS Certified Solutions Architect Study Guide – Ben Piper

• AWS for Developers – John Culkin & Mike Zazon

1. Release It! – Michael Nygard

2. Microservices Patterns – Chris Richardson

3. Building Event-Driven Microservices – Adam Bellemare

1. Observability Engineering – Charity Majors

2. The DevOps Handbook – Kim, Humble, Debois

1. Java Performance – Scott Oaks

2. Understanding the JVM – Scott Oaks (or similar advanced JVM text)

1. Fundamentals of Software Architecture – Mark Richards

2. Cloud Native Patterns – Cornelia Davis

#Assesst Creation

Games

- **“Drawing on the Right Side of the Brain” – Betty Edwards**
- **“Color and Light: A Guide for the Realist Painter” – James Gurney**
- **“Digital Art Masters” – 3DTotal Publishing**
- **“Music Theory for Computer Musicians” – Michael Hewitt**
- **“Introducing Autodesk Maya 2024” – Dariush Derakhshani (or Blender equivalent if using open source)**
- **“Polygonal Modeling: Basic and Advanced Techniques” – Mario Russo**
- **“Blender 3D: Noob to Pro” – online wikibook (optional)**
- **“Digital Sculpting with Mudbox: Essential Tools and Techniques” – Mike de la Flor / Bridgette Mongeon**

- **“ZBrush Character Creation: Advanced Digital Sculpting” – Scott Spencer**
- **“Digital Modeling” – William Vaughan**
- **“The PBR Guide: Physically Based Rendering Texturing” – Allegorithmic / Quixel**
- **“Digital Texturing & Painting” – Owen Demers**
- **“Substance Painter Projects: Learn Texturing Through Practical Exercises”**
- **“Game VFX: All-In-One Guide to Creating Effects for Games” – Alex Alvarez**
- **“Digital Compositing for Film and Video” – Steve Wright**
- **“The Art of Game Effects” – 3DTotal / Gnomon Workshop (supplementary)**
- **“Stop Staring: Facial Modeling and Animation Done Right” – Jason Osipa**
- **“Character Rigging and Advanced Animation Techniques” – George Maestri**
- **“Mocap for Game Developers” – Eric Allen**

- **“The Game Audio Tutorial: A Practical Guide to Sound and Music for Interactive Games” – Richard Stevens, Dave Raybould**
- **“Composing Music for Games” – Chance Thomas**
- **“Designing Sound” – Andy Farnell (also useful for procedural sound effects)**
- **“Game Art Complete” – Andrew Gahan**
- **“Creating Characters with 3D Studio Max and Maya” – Adam Watkins**
- **“Real-Time Rendering” – Akenine-Möller et al., chapters on textures and materials**
- **“Level of Detail for 3D Graphics” – David Luebke et al.**
- **“Game Development Essentials: Advanced 3D Game Programming” – Jeannie Novak (selected chapters)**

#Database

Management

▣ **“Database System Concepts” — Silberschatz, Korth**

▣ **“SQL and Relational Theory” — C. J. Date**

“SQL: The Complete Reference”

▣ **“Designing Data-Intensive Applications” — Martin
Kleppmann**

- **PostgreSQL: Up and Running**

- **MySQL Cookbook**

- **High Performance MySQL**

▣ **“Seven Databases in Seven Weeks”**

#Digital Forensics

**1. Computer Systems: A Programmer’s Perspective –
Bryant & O’Hallaron**

2. The C Programming Language – K&R

1. Practical Binary Analysis – Dennis Andriesse

2. File System Forensic Analysis – Brian Carrier

3. The Art of Unix Programming – Eric Raymond

(selected sections)

1. Multimedia Systems – Ralf Steinmetz & Klara Nahrstedt

3. The Data Compression Book – Mark Nelson

1. PDF Explained – John Whittington

3. ISO/IEC 14496-12 (MP4 spec)

1. The Art of Software Security Assessment – Dowd, McDonald, Schuh

2. Secure Coding in C and C++ – Robert Seacord

3. Fuzzing for Software Security Testing – Ari Takanen et al.

1. Reverse Engineering for Beginners – Dennis Yurichev

2. Practical Malware Analysis – Sikorski & Honig

1. Digital Image Forensics – Husrev Taha Sencar & Nasir Memon

2. Photo Forensics – Hany Farid

Pegasus Level Threats

1. Computer Systems: A Programmer's Perspective – Bryant & O'Hallaron

2. The C Programming Language – K&R

1. Practical Binary Analysis – Dennis Andriesse

3. The Art of Unix Programming – Eric Raymond (selected sections)

1. Multimedia Systems – Ralf Steinmetz & Klara Nahrstedt

2. Video Codec Design – Iain E. Richardson

3. The Data Compression Book – Mark Nelson

PHASE 3 — Container Formats & Metadata

1. PDF Explained – John Whittington

2. PNG: The Definitive Guide – Greg Roelofs

3. ISO/IEC 14496-12 (MP4 spec)

1. The Art of Software Security Assessment – Dowd, McDonald, Schuh

#Driver

Programming

Installable Client Driver (ICD) model

- ▣ **Windows Internals**
- ▣ **Practical Binary Analysis**
- ▣ **Windows Kernel Programming**
- ▣ **Windows Internals**
- ▣ **Practical Binary Analysis**
- ▣ **Windows Kernel Programming**
- ▣ **Windwos Internals**
- ▣ **PC Architecture**

#AAA Game Engine

- **“C++ Primer (5th Edition)” – Stanley B. Lippman, Josée Lajoie, Barbara Moo**
- **“Effective Modern C++” – Scott Meyers**
- **“C++ Concurrency in Action” – Anthony Williams**

- **“3D Math Primer for Graphics and Game Development”**
– **Fletcher Dunn, Ian Parberry**
- **“Mathematics for 3D Game Programming and Computer Graphics”** – **Eric Lengyel**
- **“OpenGL Programming Guide (Red Book)”** – **9th Edition**
- **“Introduction to 3D Game Programming with DirectX 11”** – **Frank Luna**
- **“Vulkan Programming Guide”** – **Graham Sellers, John Kessenich**
- **“Game Engine Architecture”** – **Jason Gregory**
- **“Real-Time Rendering”** – **Tomas Akenine-Möller et al.**
- **“Game Coding Complete”** – **Mike McShaffry**
- **“Programming in Lua”** – **Roberto Ierusalimschy**
- **“Game Scripting Mastery”** – **Alex Varanese**
- **“CPU/GPU Optimization for Real-Time Graphics”** –
Damien Mauger & Mike Bailey (papers/books)
- **“Game Programming Patterns”** – **Robert Nystrom**
- **“Multiplayer Game Programming: Architecting Networked Games”** – **Joshua Glazer, Sanjay Madhav**

- **“Game Physics Engine Development” – Ian Millington**
- **“Pro Game Programming” – Mike McShaffry (Tooling chapters)**
- **“Level Design: Concept, Theory, and Practice” – Rudolf Kremers / Robin Hunicke (combined readings)**
- **“Advanced Game Programming” – Gregory Wilson / Steve Rabin (selected chapters)**
- **“Real-Time Rendering 4th Edition” – again, for advanced chapters**
- **“Networked Physics” (papers and tutorials)**
- **“UML Distilled” – Martin Fowler**
- **“Applying UML and Patterns” – Craig Larman**
- **OpenGL: “OpenGL Programming Guide (Red Book)” – 9th Edition**
- **“Game Audio Programming” – Guy Somberg**
- **“Game Audio Development” – Richard Stevens, Dave Raybould**
- **Optional / Advanced: “Designing Sound” – Andy Farnell**
- **1111111**

- **Designing Sound – Andy Farnell – Procedural audio, DSP.**
- **Introducing Autodesk Maya 2024 – Dariush Derakhshani – Modeling basics, UVs, rigging.**
- **Polygonal Modeling: Basic & Advanced Techniques – Mario Russo – Topology, edge flow.**
- **Digital Sculpting with Mudbox – Mike de la Flor / Bridgette Mongeon – Sculpting for characters & props.**
- **ZBrush Character Creation – Scott Spencer – High-poly sculpting, baking for game-ready assets.**
- **Digital Modeling – William Vaughan – Artistic + technical integration.**
- **The PBR Guide – Allegorithmic / Quixel – Industry-standard physically-based materials.**
- **Digital Texturing & Painting – Owen Demers – Layered painting, stylized and realistic texturing.**
- **Substance Painter Projects – Hands-on texture creation.**
- **Game VFX – Alex Alvarez – Particles, shaders, explosions, magic.**

- **Digital Compositing for Film & Video – Steve Wright – Layering and compositing concepts.**
- **The Art of Game Effects – 3DTotal/Gnomon – AAA case studies and workflows.**
- **Stop Staring – Jason Osipa – Facial rigging & expressions.**
- **Character Rigging & Advanced Animation – George Maestri – Skeletal rigs, inverse kinematics.**
- **The Game Audio Tutorial – Stevens & Raybould – Adaptive music, FMOD/Wwise, sound design.**
- **Composing Music for Games – Chance Thomas – Themes, layering, adaptive composition.**
- **Level of Detail for 3D Graphics – David Luebke et al. – LODs, memory optimization, efficient meshes.**

#Drawing and Modelling

Books

- **“Drawing on the Right Side of the Brain” — Betty Edwards**
- **“Figure Drawing for All It’s Worth” — Andrew Loomis**
- **“Keys to Drawing” — Bert Dodson**
- **“Perspective Made Easy” — Ernest R. Norling**
- **“Force: Dynamic Life Drawing for Animators” — Michael D. Mattesi**
- **“Color and Light: A Guide for the Realist Painter” — James Gurney**
- **“Anatomy for Sculptors: Understanding the Human Figure” — Uldis Zarins & Sandis Kondrats**
- **“ImagineFX: The Ultimate Guide to Digital Painting” (various volumes)**
- **“Digital Painting Fundamentals with Corel Painter” — Wiley (or equivalent digital painting book)**
- **“The Art of XYZ” series (e.g., The Art of The Witcher 3, Death Stranding, Resident Evil)**

- **“Figure Drawing for All It’s Worth” — Andrew Loomis**
(same as above)
- **“How to Draw: Drawing and Sketching Objects and Environments from Your Imagination” — Scott Robertson & Thomas Bertling**
- **“Mastering Manga with Mark Crilley: Drawing Manga Characters and Scenes from Start to Finish” — Mark Crilley**
- **“Manga in Theory and Practice: The Craft of Creating Manga” — Hirohiko Araki**
- **“The Ghibli Art Book” / Studio Ghibli collections**
- **“Drawing Manga Art: Advanced Techniques” — Hosoi Aya & Line-of-Action style texts**
- **“Figure Drawing: Design and Expression (Stylized)”**
(various authors/publishers)
- **“Kentaro Miura Berserk Artbooks” (official artbooks)**
- **“Introducing Autodesk Maya 202x: A Comprehensive Guide” — (various authors)**
- **“Blender For Absolute Beginners” — (various authors)**

- **“Stop Staring: Facial Modeling and Animation Done Right” — Jason Osipa**
- **“Digital Modeling” — William Vaughan**
- **“ZBrush Character Sculpting” — (various authors)**
- **“The Art of Sculpting with ZBrush” — (various authors)**
- **“3D Textures and Modeling” — (various artists / handbook series)**
- **“The Artist’s Guide to Creating a 3D Animated Character” — 3D Total Publishing**
- **“Game Engine Architecture” — Jason Gregory (this is more technical but excellent for understanding real game pipelines and how art integrates with game systems)**
- **“The Art and Science of Drawing / Concepts in 3D Pipelines” (Gnomon workshop books)**
- **“The Animator’s Survival Kit” — Richard Williams**
- **“Animation From Pencils to Pixels” — Tony White**
- **“Timing for Animation” — Harold Whitaker & John Halas**

- **“Anime Mania: How to Draw Characters for Japanese Animation” — Christopher Hart**
- **“Creating Characters with Personality” — Tom Bancroft**
- **“Directing Animation” — David B. Levy**
- **“The Illusion of Life: Disney Animation” — Frank Thomas & Ollie Johnston**
- **“Producing Animation” — Catherine Winder & Zahra Dowlatabadi**
- **“Storyboard Storytelling: Visual Storytelling in Animation” — Mark Simon**
- **“Framed Ink: Drawing and Composition for Visual Storytellers” — Marcos Mateu-Mestre**

1) Digital Art & Concept Art (Realistic, AAA Game/Film Level)

- **ImagineFX (magazine & workshop compilations) — not a book but excellent hands-on guided content**
- **Concept Art & Illustration Courses – CG Spectrum**

- **Concept Art School (CGMA) – 2D Drawing, Dynamic Sketching, Environment Design, and visual storytelling courses taught by industry professionals (mentored and structured).**
- **Drawing on the Right Side of the Brain — foundational sketch skills**
- **Figure Drawing for All It's Worth — structure + anatomy**
- **How to Draw: Drawing and Sketching Objects and Environments from Your Imagination — Scott Robertson**
- **Mastering Manga with Mark Crilley — beginner to intermediate manga skills**
- **Manga in Theory and Practice — craft & storytelling insight**
- **Manga Art School: The Complete Anime & Manga Drawing Course (Udemy)**
- **Anime Art Academy Lessons**
- **The Animator's Survival Kit — Richard Williams (essential animation principles)**
- **Animation From Pencils to Pixels — Tony White**
- **Timing for Animation — Harold Whitaker & John Halas**

- **Directing Animation — David Levy**
- **Storyboard Storytelling — Mark Simon**
- **Animation fundamentals courses on sites like Skillshare, Udemy, and Domestika (basic movement, squash-stretch, timing, walk cycles).**
- **Anime-style bundles on Creative Courses Group (focused beginner anime character drawing + digital art techniques)**
- **Animation and storyboarding courses through Elite Animation Academy (includes digital painting, storyboarding, character animation, and foundational 2D skills)**
- **Blender For Absolute Beginners — (various authors; practical modeling with free Blender)**
- **Introducing Autodesk Maya — official comprehensive modeling guide**
- **Digital Modeling — William Vaughan (core modeling principles)**
- **ZBrush Character Sculpting — detailed, project-based sculpting workflows**

- **The Artist's Guide to Creating a 3D Animated Character — pipeline from model → rig → texture → render**
- **Relevant 3D art & concept pipelines (Gnomon/3D Total publishing series)**
- **Blender Fundamentals (e.g., BlenderGuru, CG Cookie) — hands-on tutorials growing into advanced projects (very practical).**
- **Maya Modeling & Animation Courses (Udemy) — beginner to advanced, real 3D workflows (rigging, animation, texturing).**

Drawing Courses

- **Proko — Drawing Basics**
- **Proko — Figure Drawing / Gesture**
- **Proko — Portrait / Head Drawing**
- **Marshall Vandruff — Perspective Drawing, 1994**

Lectures

- **Drawabox — Lesson 1 onward**
- **Marc Brunet — Beginner's Guide to Drawing**

- **Sinix Design — Anatomy / Drawing / Painting**

- **Marco Bucci — Painting Fundamentals**

- **Krita — Official YouTube tutorials**

Proko → Drawabox → Marshall Vandruff → Proko

Figure/Portrait → Sinix → Marco Bucci → AAA concept-art studies.

Zbrush

- **ZBrush Beginner's Course: Sculpting Crile the**

Crocodile

- **ZBrush Hard Surface Sculpting for Beginners**

- **Character Creation for Games Vol. 1: Sculpting in**

ZBrush

- **Female Anatomy Sculpting in ZBrush**

- **Advance ZBrush Character Creation**

- **Clothes and Folds Exercises Course**

- **Creating Props for the VFX Pipeline with Maya, ZBrush**

& Substance

#Practical Graphics

API Programming

Vulkan Programming Guide

Vulkan Tutorial

Introduction to 3D Game Programming with DirectX12

#Electronics

- **Code: The Hidden Language of Computer Hardware and Software — Charles Petzold — great intuition on how hardware and code interact from the ground up.**
- **The Art of Electronics — Paul Horowitz & Winfield Hill — the gold standard reference for practical electronics.**
- **Solid State Electronic Devices — Ben G. Streetman — widely used in university courses on semiconductor fundamentals.**

- **Semiconductor Device Fundamentals — Robert F. Pierret — another excellent physics-first text.**
- **Digital Design and Computer Architecture — David Harris & Sarah Harris — bridges logic design with real CPU architecture.**
- **Digital Design — M. Morris Mano — classic, beginner-friendly logic design.**
- **Computer Organization and Design (e.g., ARM/RISC-V edition) — Patterson & Hennessy — industry standard for how processors really work.**
- **Computer Architecture: A Quantitative Approach — Patterson & Hennessy — advanced, industry-level architecture.**
- **Making Embedded Systems — Elecia White — practical intro to embedded development.**
- **Embedded Systems Design — Arnold Berger — full cycle embedded design (hardware + tools).**
- **Programming Embedded Systems in C and C++ — Michael Barr & Andy Oram — solid embedded SW foundations.**

- **Embedded Systems with ARM Cortex-M**

Microcontrollers in Assembly Language and C — Yifeng Zhu — modern MCU programming.

- **ARM Assembly Language: Fundamentals and Techniques — William Hohl — deep ARM knowledge.**

- **The Verilog® Hardware Description Language — Donald Thomas & Philip Moorby — excellent intro to Verilog HDL.**

- **FPGA Prototyping by VHDL/Verilog Examples — Pong P. Chu — bottom-up digital hardware design practice.**

- **CMOS VLSI Design: A Circuits and Systems Perspective — Neil Weste & David Harris — canonical for chip design fundamentals.**

- **Digital Integrated Circuits: A Design Perspective — Jan Rabaey, Anantha Chandrakasan, Borivoje Nikolic — broad, deep digital IC design.**

- **Modern VLSI Design: Systems on Silicon — Wayne Wolf — advanced SoC & integration practice.**

- **Algorithms for VLSI Physical Design Automation — Naveed Sherwani — algorithms and CAD techniques used in IC layout.**
- **Practice problem books (e.g., for VLSI interview prep) — Shahin Nazarian — good reinforcement.**
- **Chip War: The Fight for the World's Most Critical Technology — Chris Miller — strategic and economic context for the semiconductor industry, not technical but useful.**
- **Physics & Electronics: Streetman + Pierret + Horowitz.**
- **Digital Logic: Mano + Harris & Harris.**
- **Computer Architecture: Patterson & Hennessy.**
- **Embedded Systems: White + Barr & Oram.**
- **HDL & FPGA: Chu + Verilog language text.**
- **IC / VLSI: Weste & Harris + Rabaey + Sherwani.**
- **Practical Projects: apply on real boards (STM32/FPGA) and open toolchains.**

#Hashing

5 The Joy of Cryptography

“Rosulek's ‘Joy of Cryptography’ will always be my first recommendation.”

- **Serious Cryptography**
- **Understanding Cryptography**
- **Cryptography Engineering**
- **Applied Cryptography**
- **Real-World Cryptography**
- **Signal Protocol / Double Ratchet docs**
- **Practical Cryptography in C/C++**

#Pegasus Malware Analyst

- **Practical Malware Analysis — your starting point**
- **Reversing: Secrets of Reverse Engineering (Eldad Eilam) — solid coverage of reverse engineering fundamentals and malware context**

- **Mastering Malware Analysis** — goes deeper into anti-analysis, evasion, cross-platform threats
- **Malware Analysis and Detection Engineering** — modern, practical coverage of static, dynamic, and detection engineering — great bridge to real-world detection workflows
- **The Art of Memory Forensics** — critical if you want to analyze live systems or extracted memory
- **Windows Internals** — essential for understanding system behavior at a level Pegasus exploits often target
- **The Ghidra Book (Definitive Guide)** — strong static analysis foundation

#Encodings

- 1 **Unicode Demystified**
- 2 **The Unicode Standard**
- 3 **Design Patterns : Elements of Reusable Object Oriented Software**
- 4 **Algorithms on Strings , Trees, and Sequences**

5 The Craft of Text Editing

6 Compilers: Principles , Techniques and tools

7 Fonts and Encoding

#AI

**** “Artificial Intelligence: A Modern Approach” — Stuart Russell & Peter Norvig**

**** “Mathematics for Machine Learning” — Deisenroth, Faisal & Ong**

**** “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow” — Aurélien Géron**

**** “The Hundred-Page Machine Learning Book” — Andriy Burkov**

**** “Deep Learning” — Ian Goodfellow, Yoshua Bengio & Aaron Courville**

**** “Pattern Recognition and Machine Learning” — Christopher M. Bishop**

**** “Machine Learning: A Probabilistic Perspective” — Kevin P. Murphy**

**** “Reinforcement Learning: An Introduction” — Sutton & Barto**

**** “Artificial Intelligence for Games” — Ian Millington & John Funge**

**** “Programming Game AI by Example” — Mat Buckland**

**** “Behavior Trees in Robotics and AI” — A Practical Introduction**

**** “The Master Algorithm” — Pedro Domingos**

**** “You Look Like a Thing and I Love You” — Janelle Shane**

• Introduction to algorithms

Numerical Recipes in C

Numerical Linear Algebra

Matrix Computations

• Effective Modern C++

• C++ Concurrency in Action

• Optimized C++

Machine Learning Systems

Designing Machine Learning Systems

- **“Deep Learning for Coders with fastai and PyTorch”**
(Francois Chollet / Jeremy Howard)
- **“Transformers from Scratch” blog / GitHub repos**
- **“CUDA by Example” (Jason Sanders, Edward Kandrot)**

#Job

C++

- **Programming: Principles and Practice Using C++ – Bjarne Stroustrup**
- **Effective Modern C++ – Scott Meyers**
- **The Linux Programming Interface – Michael Kerrisk**
- **Beej's Guide to Network Programming (free)**
- **TCP/IP Illustrated, Volume 1**
- **Learning SQL by Alan Beaulieu**

Java

- **Head First Java — Kathy Sierra, Bert Bates**
- **Effective Java — Joshua Bloch**
- **Learning SQL — Alan Beaulieu**

- **Spring Start Here — Laurentiu Spilca**
- **Spring in Action — Craig Walls**
- **Web API Design (Brian Mulloy)**
- **Clean Code — Robert C. Martin**
- **Head First Design Patterns**
- **Clean Architecture — Robert C. Martin**
- **Designing Data-Intensive Applications — Martin Kleppmann**

- **Computer Systems: A Programmer's Perspective (CS:APP)**
- **The Linux Command Line — William Shotts**
- **The Linux Programming Interface — Michael Kerrisk**
- **Maven: The Definitive Guide — Sonatype**
- **Learning SQL — Alan Beaulieu**
- **PostgreSQL: Up and Running — Regina Obe, Leo Hsu**

Book: 11. HTTP: The Definitive Guide

Book: 12. Docker Deep Dive — Nigel Poulton

- **Java How to Program — Paul Deitel, Harvey Deitel**
- **Data Structures and Algorithms in Java — Robert Lafore**
- **Docker Deep Dive — Nigel Poulton**
- **JUnit in Action — Catalin Tudose**

#ABI

1. Linkers & Loaders — John R. Levine

2. Practical Binary Analysis — Dennis Andriesse

**3. Programming Applications for Microsoft Windows —
Charles Petzold**

4. Windows via C/C++ — Jeffrey Richter

**5. Undocumented Windows 2000 Secrets — Sven B.
Schreiber**

6. Microsoft PE/COFF Specification

7. System V AMD64 ABI Specification

8. Itanium C++ ABI

#Disassembler

- **Computer Systems: A Programmer's Perspective**
- **Programming from the Ground Up**
- **Linkers and Loaders**
- **Windows Internals**
- **Reverse Engineering for Beginners**
- **Practical Reverse Engineering**
- **Compilers: Principles, Techniques, and Tools**
- **Engineering a Compiler**
- **Advanced Compiler Design and Implementation**
- **Binary Analysis Cookbook**
- **The Art of Assembly Language**

Unreal and Technical Art/Graphics Programming Technical Artists/RealTime VFX

- **The Book of Shaders — Patricio González Vivo & Jen Lowe**
- **Unreal Engine 5 Materials Documentation**
- **Unreal Engine Material Expression Reference**

- **Real-Time Rendering — Akenine-Möller, Haines & Hoffman**

- **The Art of Code — YouTube channel**

GPU Gems 1 — NVIDIA

GPU Gems 2 — NVIDIA

GPU Gems 3 — NVIDIA

3D Math Primer for Graphics and Game Development — Fletcher Dunn & Ian Parberry

Image Processing — study this as a subject after the above, especially filtering, convolution, blur, color manipulation, and image-space operations

Procedural Generation — especially noise, fractals, SDFs, domain warping, and procedural textures

Real-Time VFX — particles, materials, distortion, animation, compositing, and stylized effects

GPU Architecture & Optimization — understand how shaders actually execute on the GPU and how to make them fast

Foundational Android Understanding

Embedded Android — Karim Yaghmour

**Android Internals: A Confectioner's Cookbook —
Jonathan Levin**

Android Security Internals — Nikolay Elenkov

**The Linux Programming Interface — Michael Kerrisk (if
you don't already have it in your Linux section)**

**ARM Architecture Reference Manual (reference, not a
beginner book)**

Image and Video Processing

- **Introduction to Data Compression (Sayood)**
- **The Data Compression Book (Nelson & Gailly)**
- **Data Compression: The Complete Reference (Salomon)**
- **PNG: The Definitive Guide**
- **Study the zlib source code after learning the theory.**

That combination gives you both the mathematical foundation and the implementation knowledge needed to eventually write your own compressors, image loaders, and asset pipeline for a game engine.

XML

**XML in a Nutshell — Elliotte Rusty Harold & W. Scott
Means**

XML Pocket Reference — Simon St. Laurent

XAML

XAML in a Nutshell — Lorn Potter